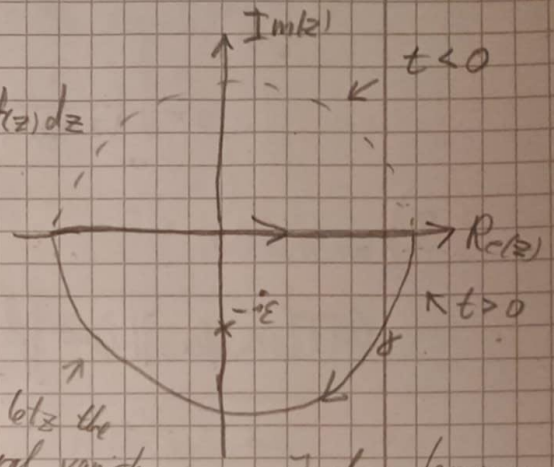
9.2)

assuming $t>0$:

time part.

show: $\Theta(t) = i\int_{-\infty}^{\infty} \frac{1}{2\pi} \frac{e^{-izt}}{z+i\epsilon}\, dz = i\int_{-\infty}^{\infty} f(z)\, dz$



simple pole at $z = -i\epsilon$

$$\int_{-\infty}^{\infty} f(x)\,dx + I_2 = \oint f(z)\,dz = -2\pi i \sum_j \mathrm{Res}(f(z_j))$$

e^{-izt} lets the integral vanish for the lower loop

Jordan's lemma

$$I_2 = \lim_{R\to\infty} \int_\gamma e^{-izt} Q(z)\,dz = 0$$

$\mathrm{Res}(f(-i\epsilon)) =$

$$\mathrm{Res}(f(z)) = \lim_{z\to z_0} \left((z-z_0) f(z) \right)$$

$$\mathrm{Res}(f(-i\epsilon)) = \lim_{z\to -i\epsilon} \left(\cancel{(z+i\epsilon)} \frac{e^{-izt}}{\cancel{(z+i\epsilon)}} \right) = e^{-\epsilon t}$$

for $\epsilon \to 0^+$

$$\Rightarrow \Theta(t) = \frac{i}{2\pi} \cdot \int_{-\infty}^{\infty} f(z)\,dz = \frac{i}{\cancel{2\pi}} \cdot (-\cancel{2\pi} i)\, e^{-\epsilon t} = e^{-\epsilon t} \to 1$$

~~for $t>0$~~

$$\Rightarrow \frac{i}{2\pi} \int \frac{e^{-izt}}{z+i\epsilon}\, dz = \begin{cases} 1, & \text{for } t>0 \\ 0, & \text{for } t<0 \end{cases} = \Theta(t)$$

show now: ~~$\Theta(t) e^{-i\omega_p t}$~~

$$\Theta(t) = i\int_{-\infty}^{\infty} \frac{dz}{(2\pi)} \frac{e^{-izt}}{z+i\epsilon}$$

← basically a fourier integral

it transforms $\Theta(z) \to \Theta(t)$

$$\Theta(z) = \frac{i}{z+i\epsilon}$$

$$\Theta(t) = \mathcal{F}\left[\frac{i}{z+i\epsilon}\right]$$

shift theorem: $\mathcal{F}[\Theta(z-\omega_p)] = \Theta(t)\, e^{-i\omega_p t}$

$$\mathcal{F}[\Theta(z-\omega_p)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\frac{i}{z-\omega_p+i\epsilon} \right) e^{-izt}\, dz \quad , \quad z \to p_0 = E$$